RESTRICTION OF SCALARS IS LEFT ADJOINT TO HOM FROM S

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof ends with the author's sentence that the verification of the two constructions being mutually inverse is omitted; that omission is routine, the two constructions themselves are fully written, and the ending sentence is kept verbatim inside the span.

Editorial note. The counted spans contain no \ref and no \cite, so the card carries no navigation stub and no bibliography.

Allowed standard prerequisites (editorial).

- The R-module structure on Hom from S. Used implicitly at the construction of alpha' in the counted proof (lines 2720-2722). Makes Hom_R(S, M) an S-module, which is required for the target of the adjunction map to be an S-module map.
- Multiplication by an element of S inside N. Used at proof line 2722 in the formula alpha'(n)(s) = alpha(s n). Supplies the S-action that the bijection has to respect.

Lemma 0.1. *Let $R \to S$ be a ring map. The functors $Mod_S \to Mod_R$, $N \mapsto N_R$ (restriction) and $Mod_R \to Mod_S$, $M \mapsto \mathrm{Hom}_R(S, M)$ are adjoint functors. In a formula*

$$\mathrm{Hom}_R(N_R, M) = \mathrm{Hom}_S(N, \mathrm{Hom}_R(S, M))$$

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Proof. If $\alpha : N_R \to M$ is an R-module map, then we define $\alpha' : N \to \mathrm{Hom}_R(S, M)$ by the rule $\alpha'(n) = (s \mapsto \alpha(sn))$. If $\beta : N \to \mathrm{Hom}_R(S, M)$ is an S-module map, we define $\beta' : N_R \to M$ by the rule $\beta'(n) = \beta(n)(1)$. We omit the verification that these constructions are mutually inverse. $\square$